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(54) **FASTENER APPLICATOR WITH INTERLOCK**

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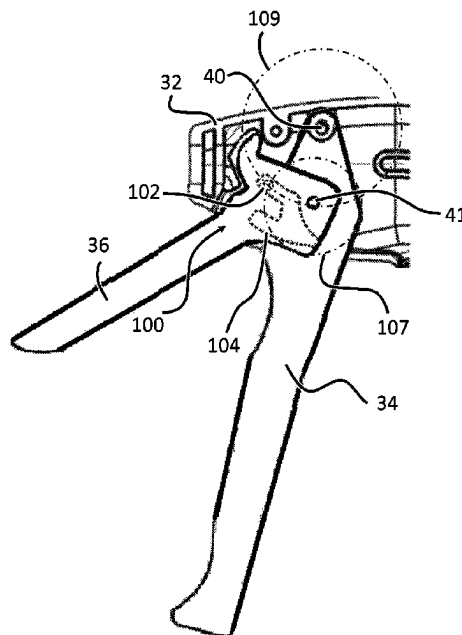
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(57) **ABSTRACT**
The present disclosure relates to a fastener applicator including a handle, a first trigger coupled rotatably with respect to the handle, and a second trigger coupled rotatably with respect to the handle. An interlock is disposed with the second trigger and comprises a cam follower engaged in a cam channel that selectively prevents rotation of the second trigger relative to the first trigger depending on a location of the cam follower within the cam channel. The location of the cam follower in the cam channel is set by a position of the first trigger. Methods of operating the various embodiments disclosed are also provided.

6 Claims, 14 Drawing Sheets



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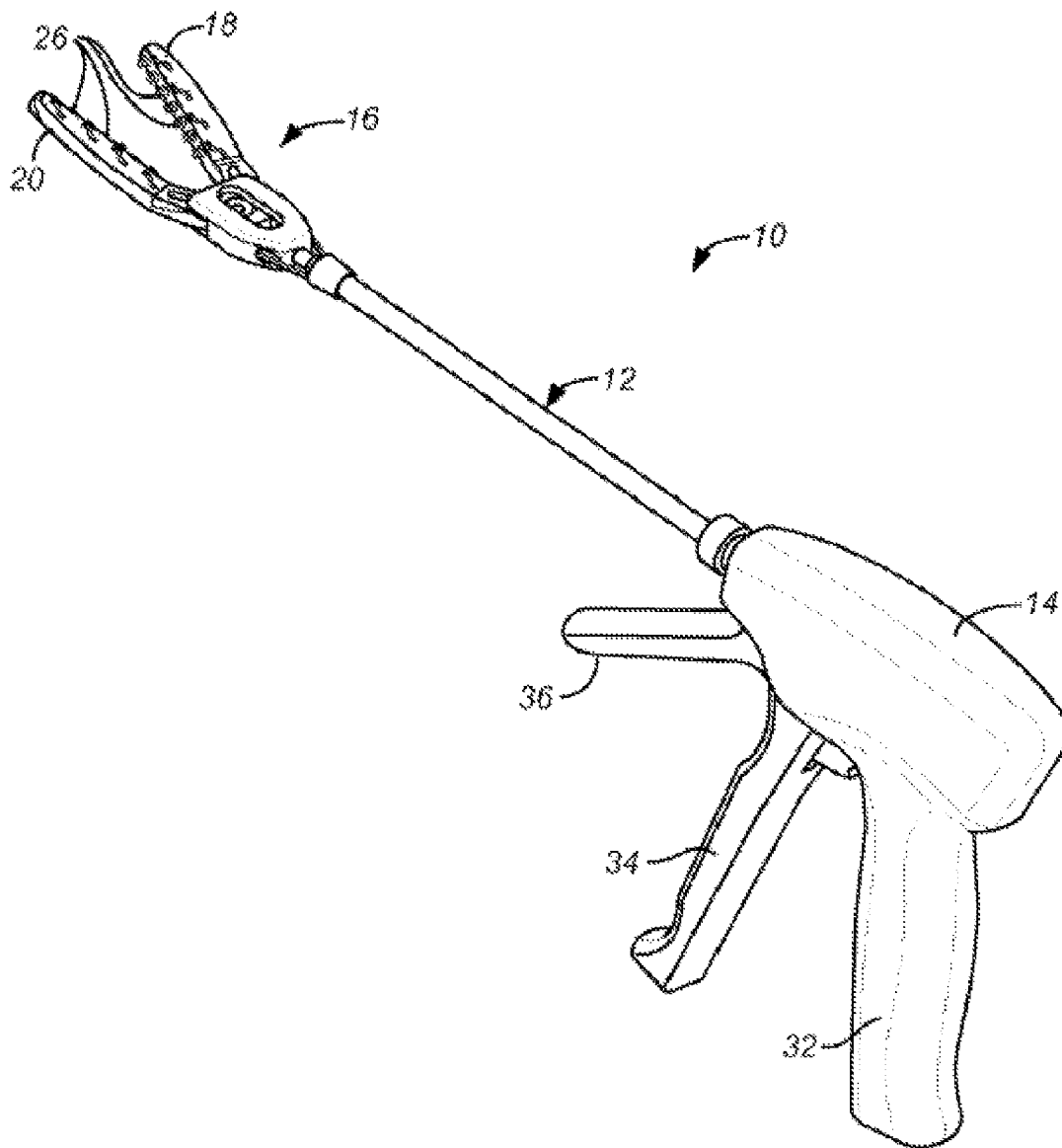


FIG. 1

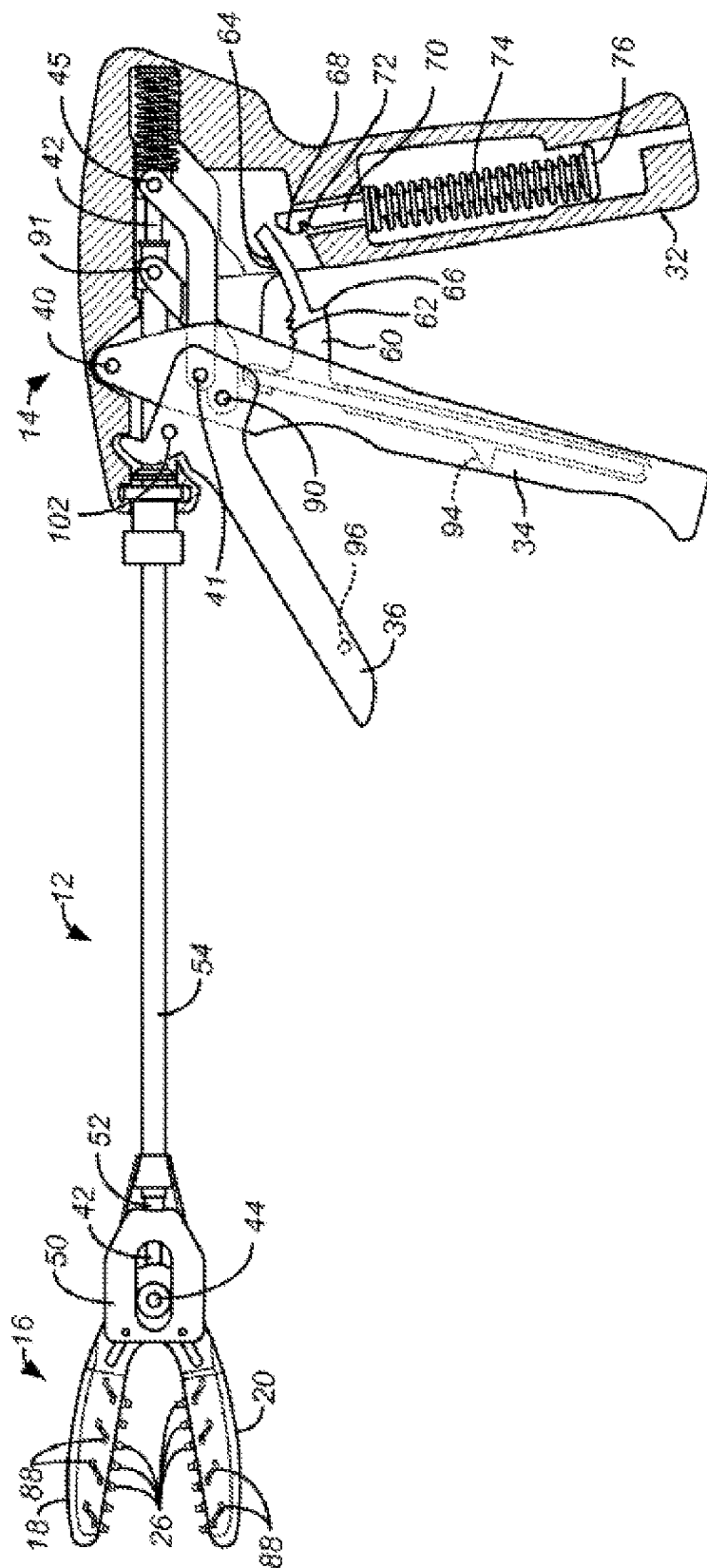


FIG. 2

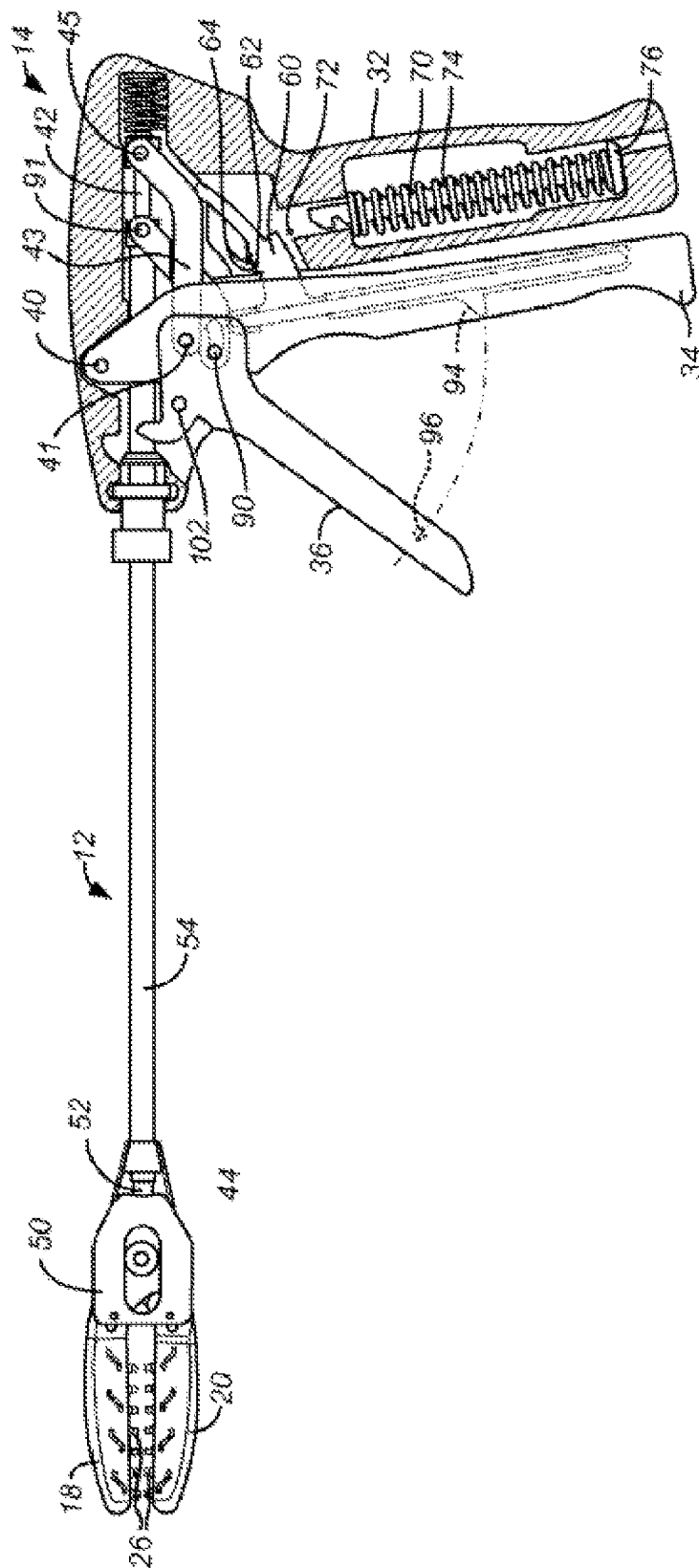


FIG. 3

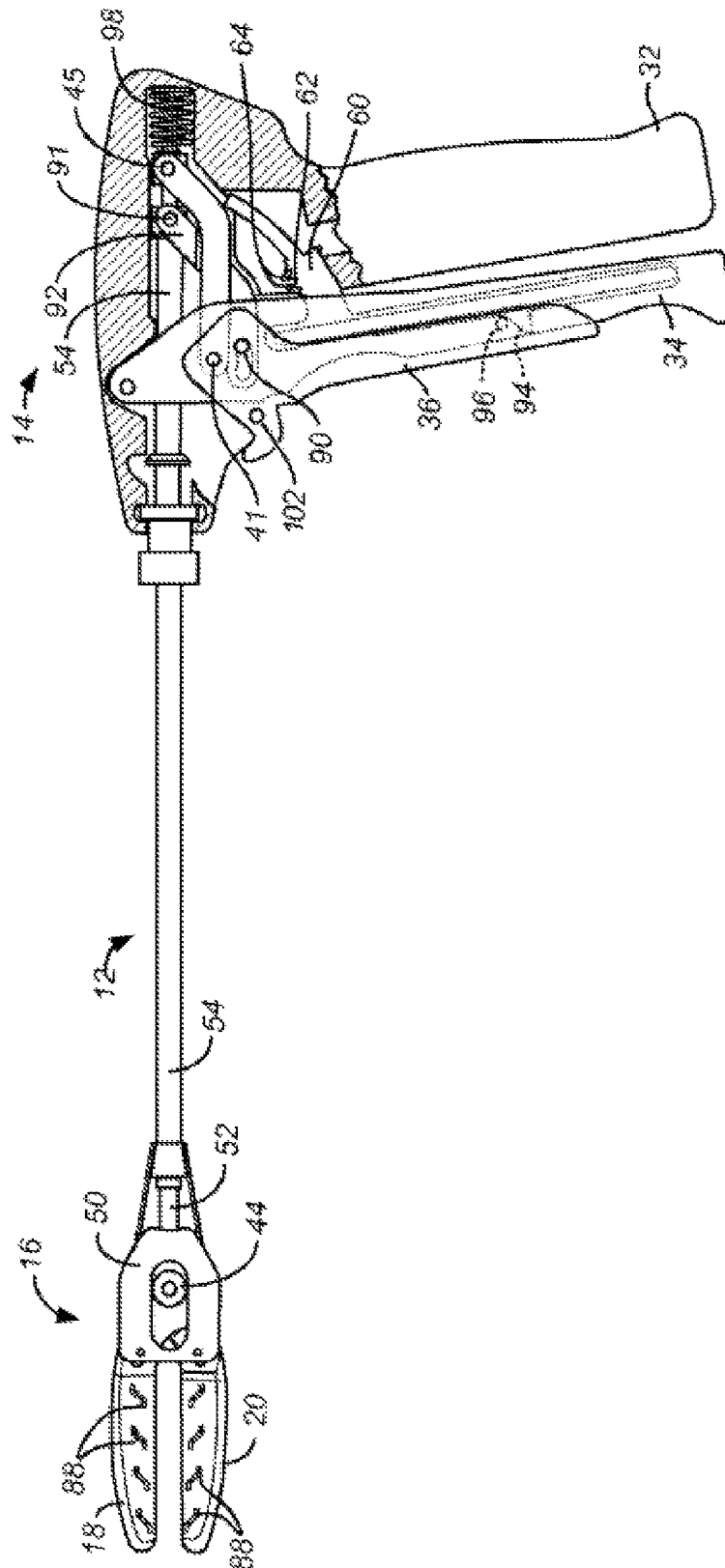


FIG. 4

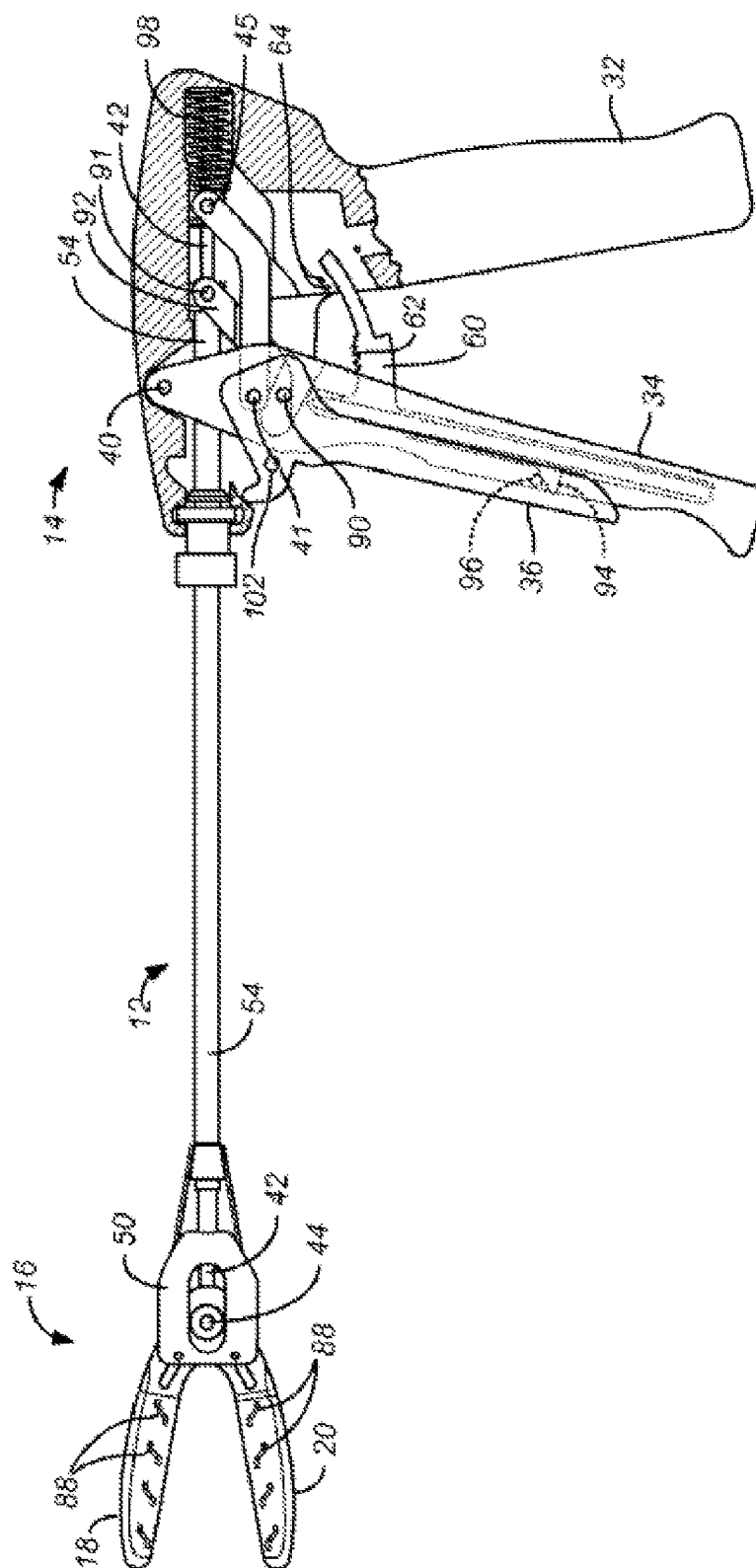


FIG. 5

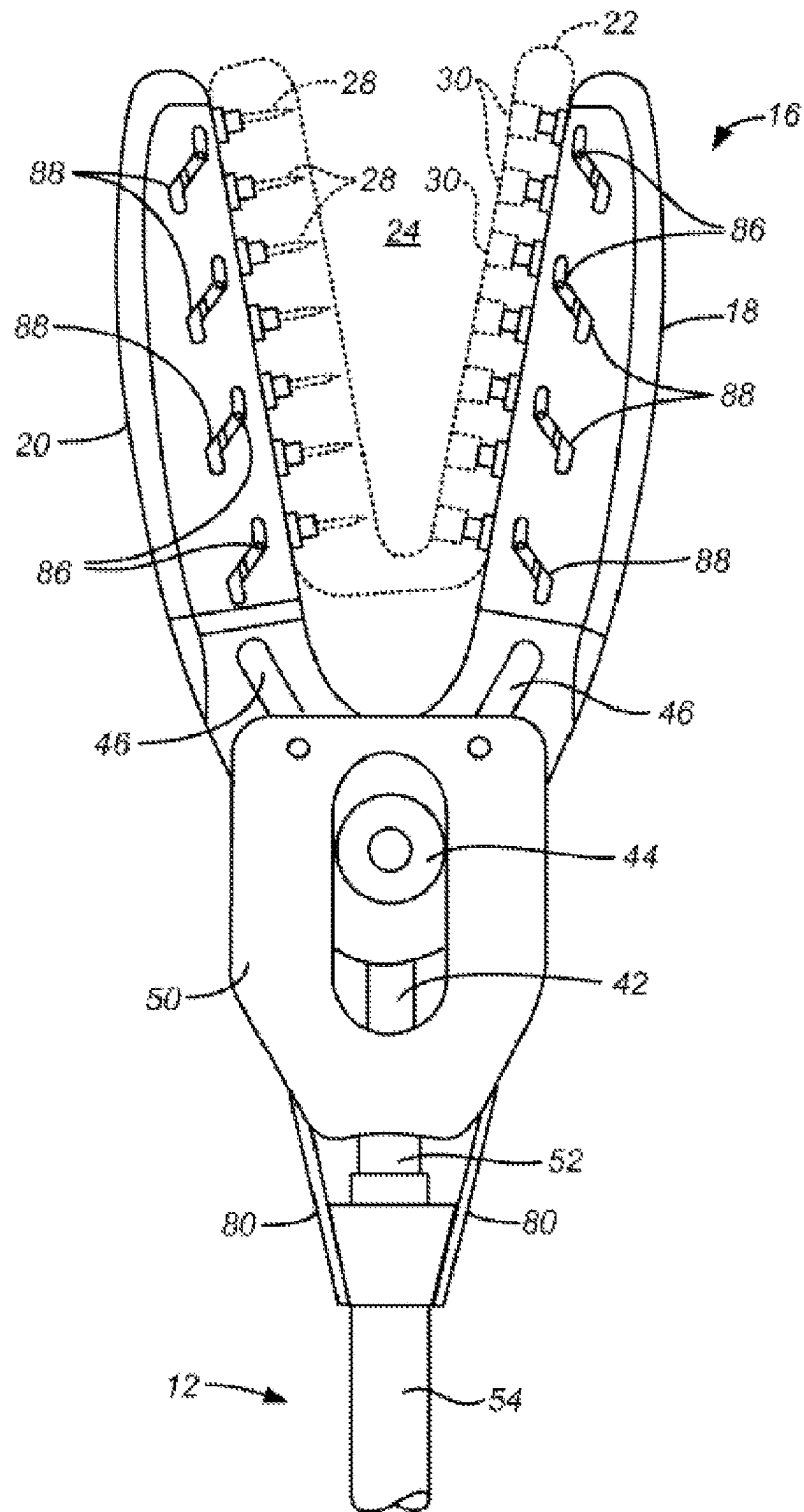


FIG. 6A

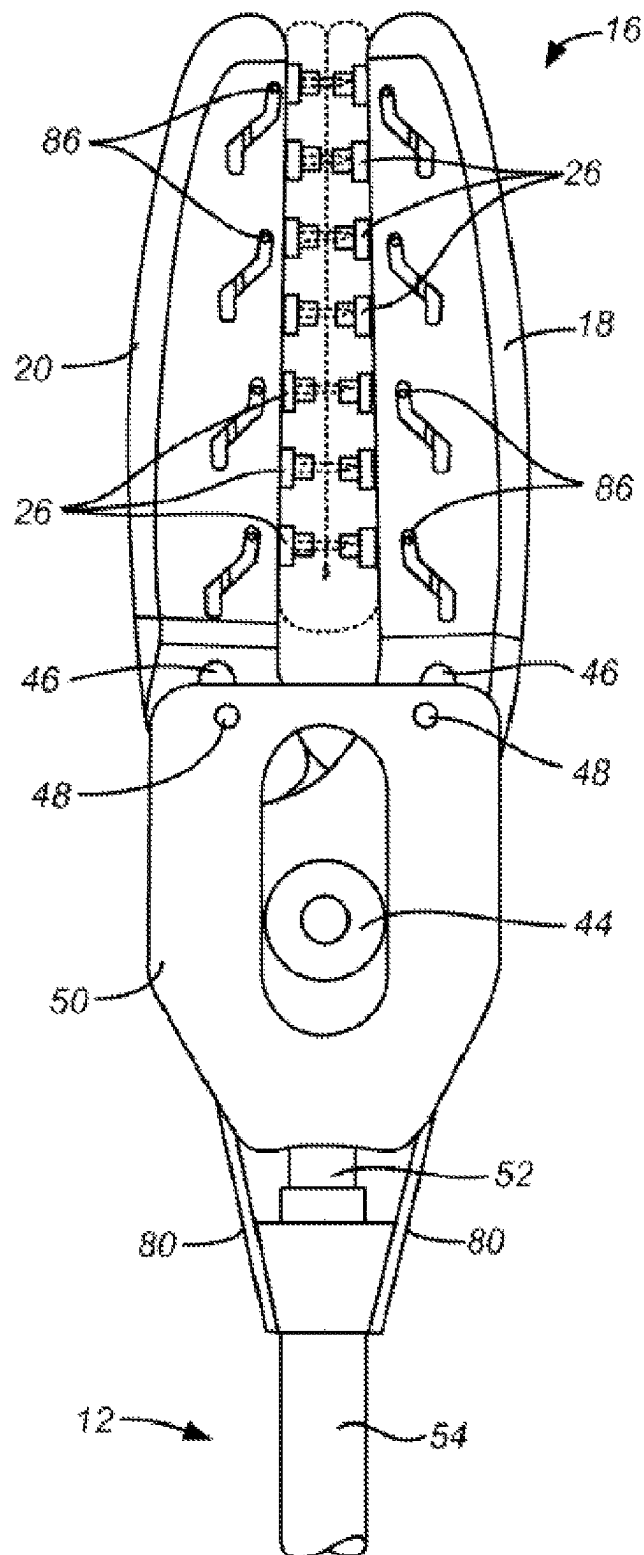


FIG. 6B

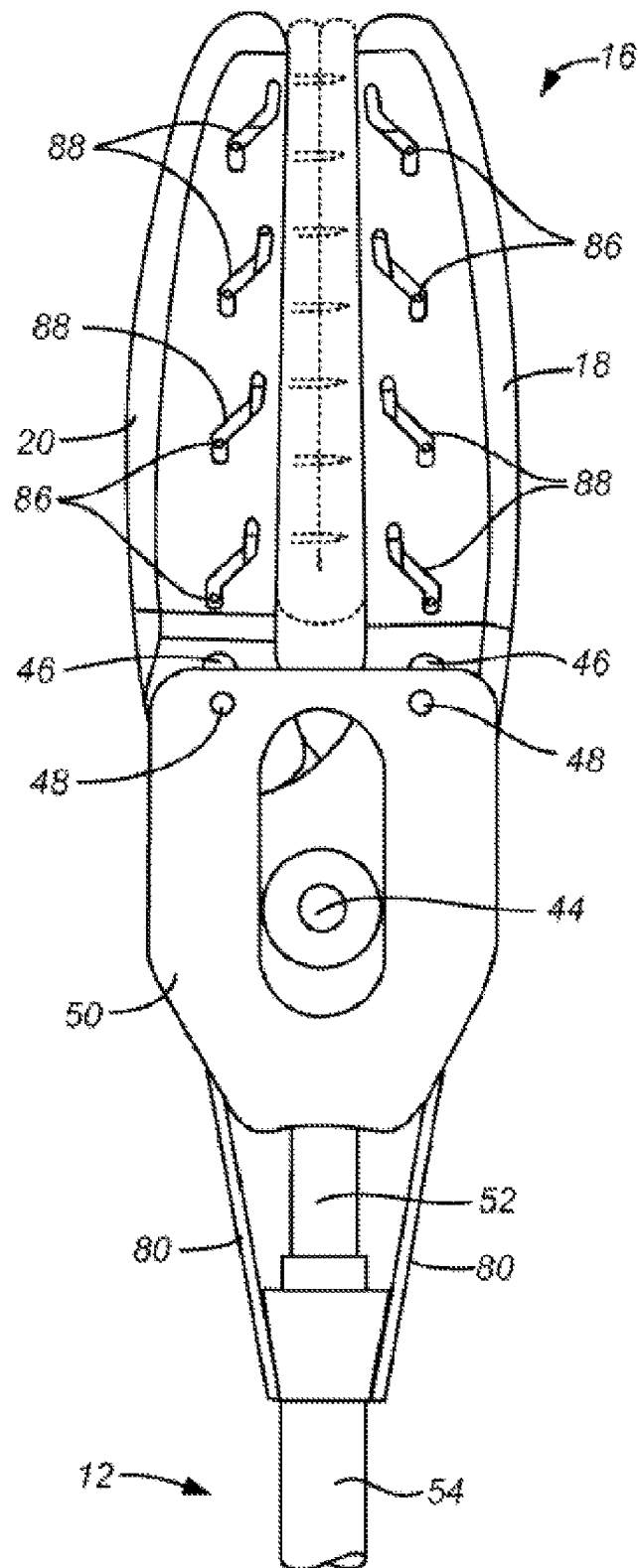


FIG. 6C

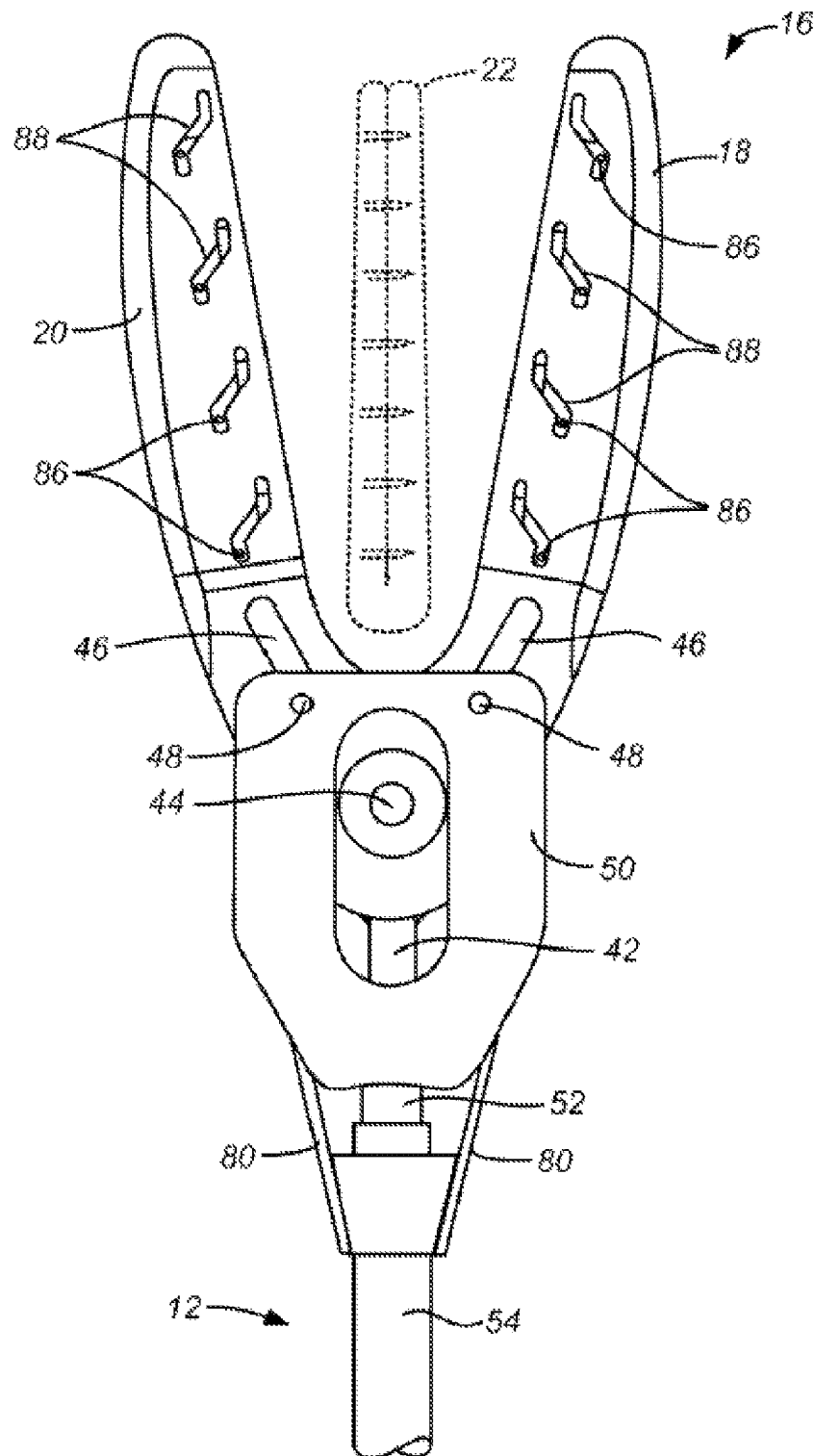


FIG. 6D

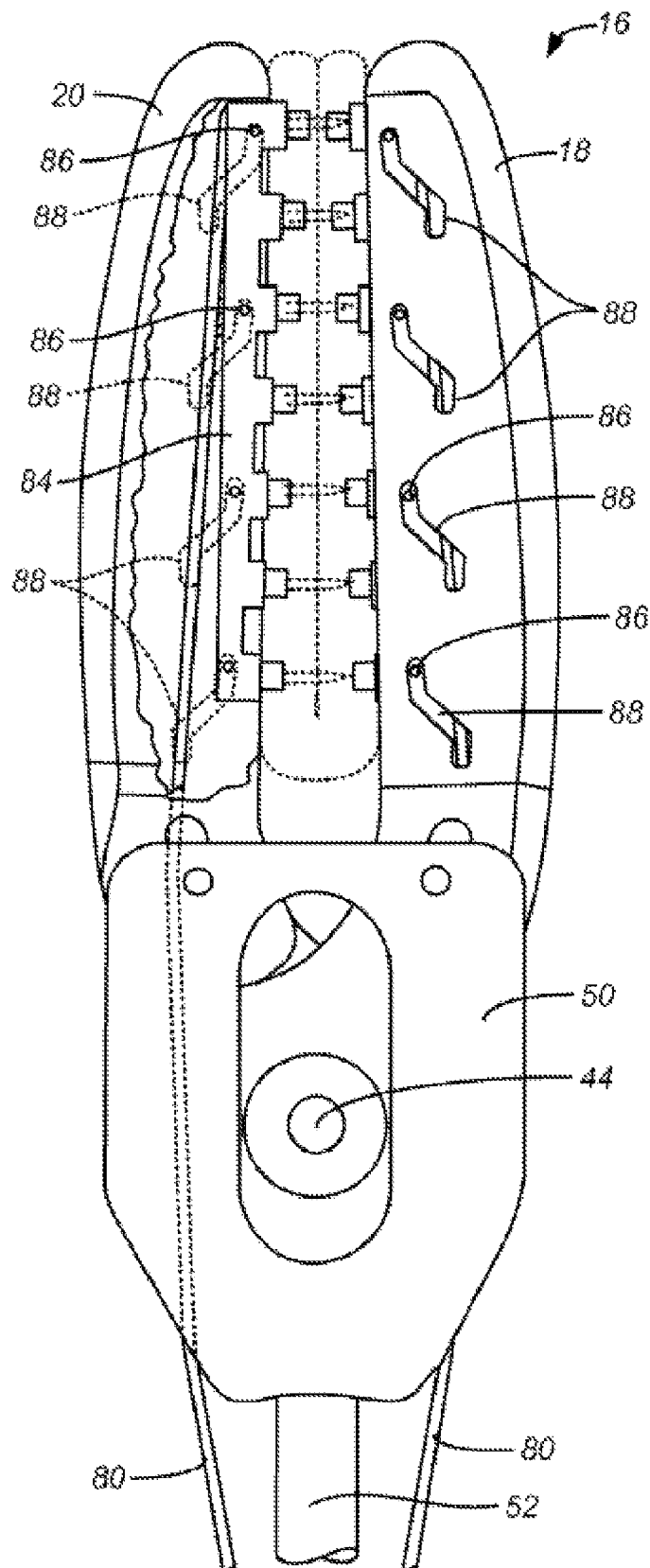
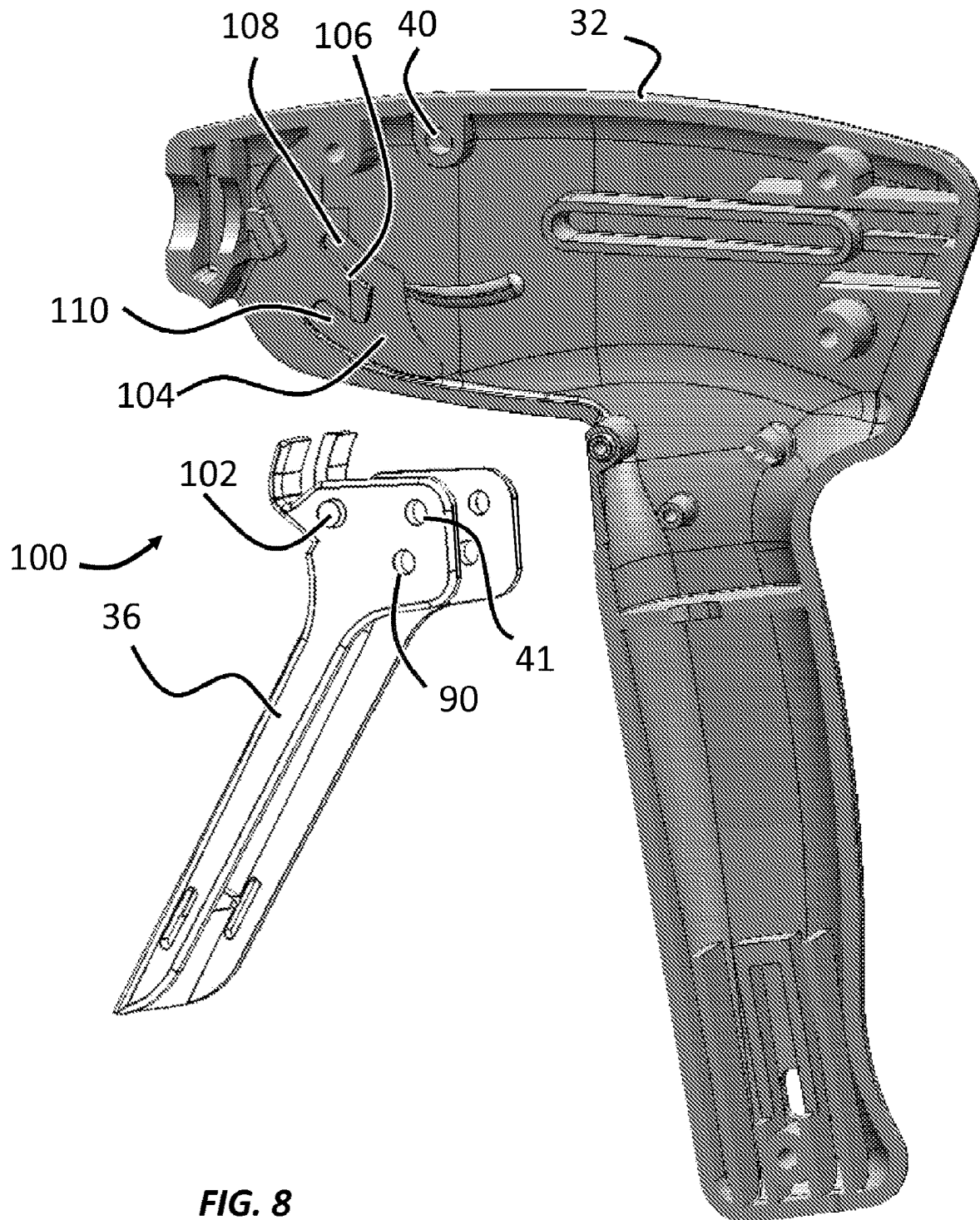
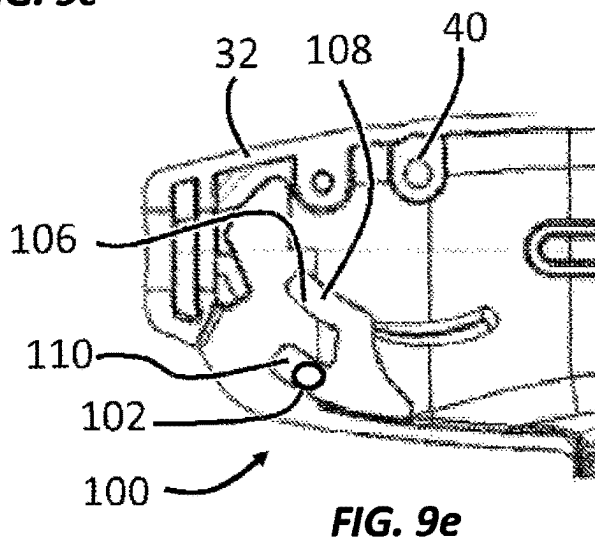
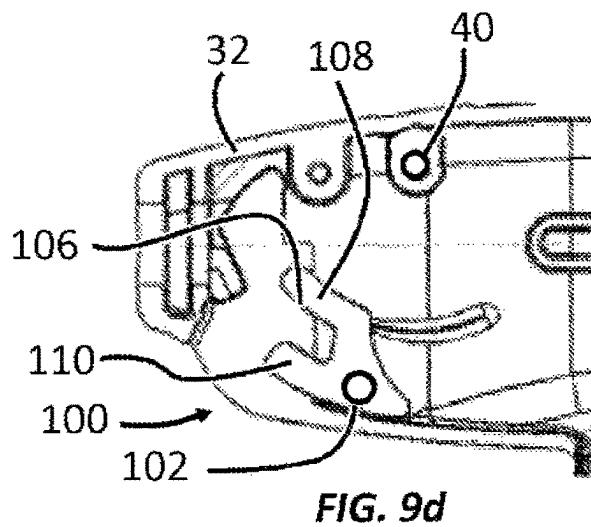
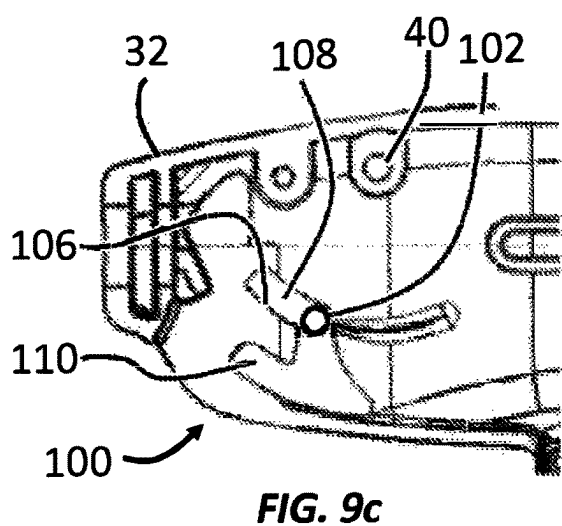
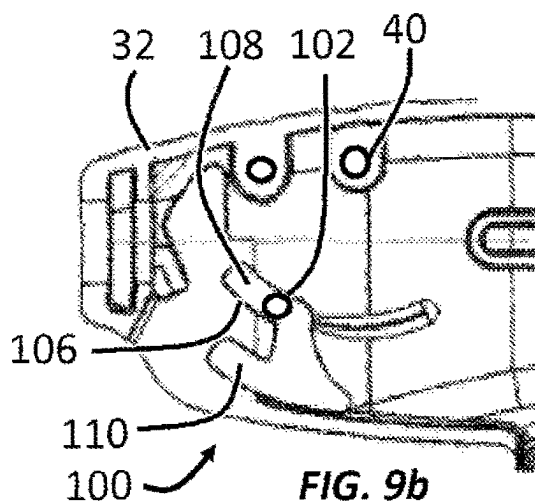
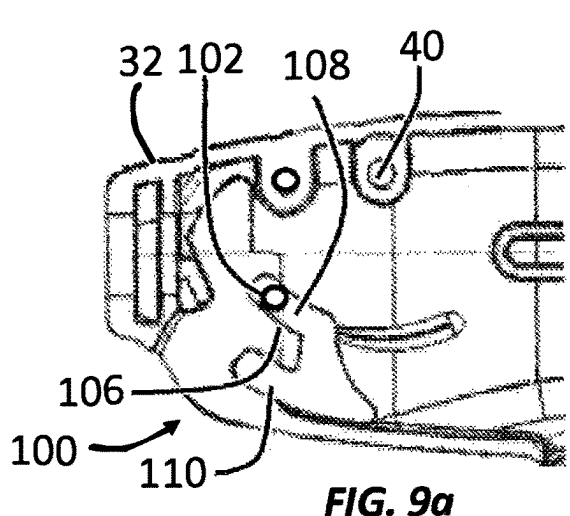
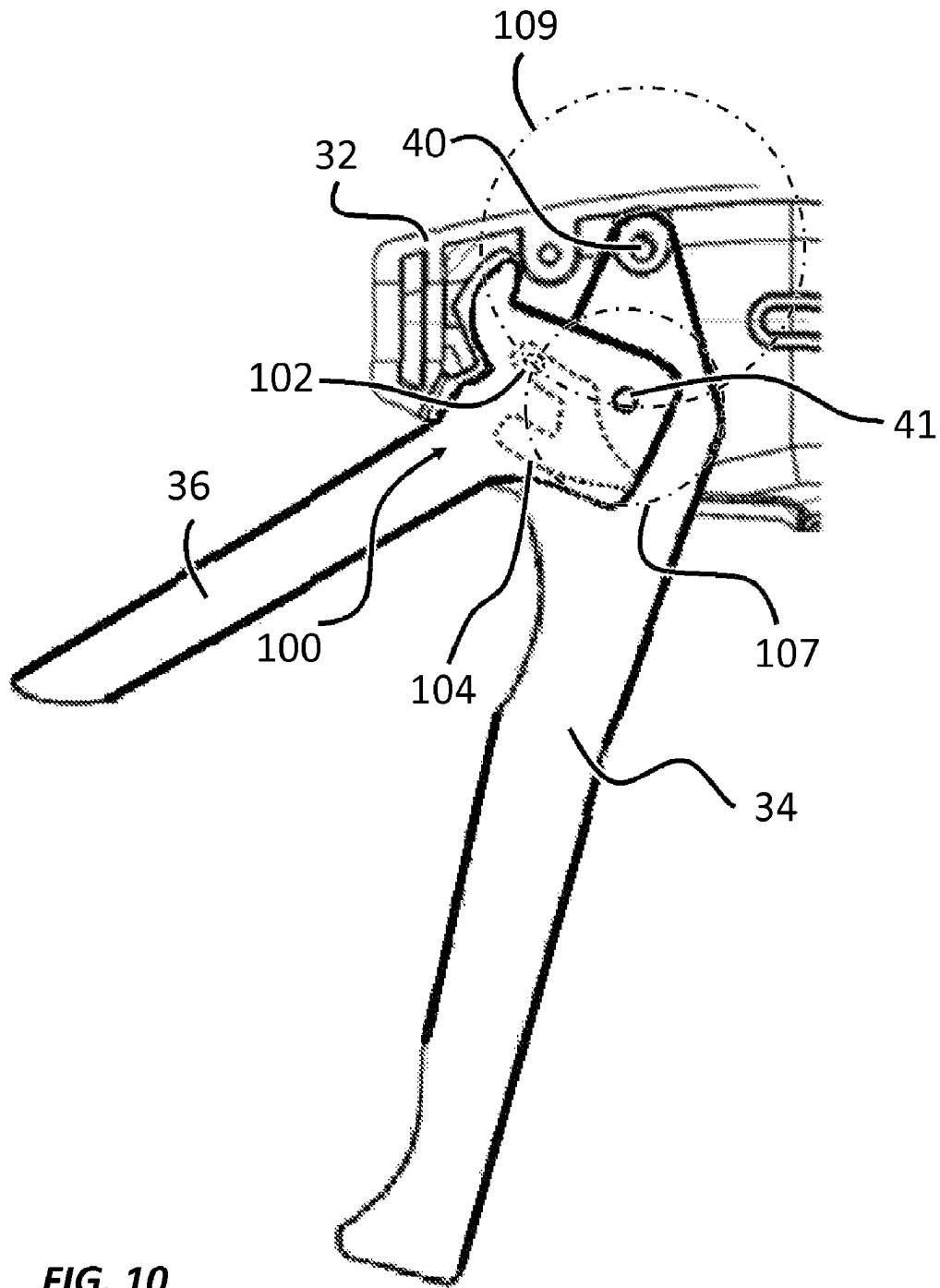


FIG. 7

**FIG. 8**



**FIG. 10**

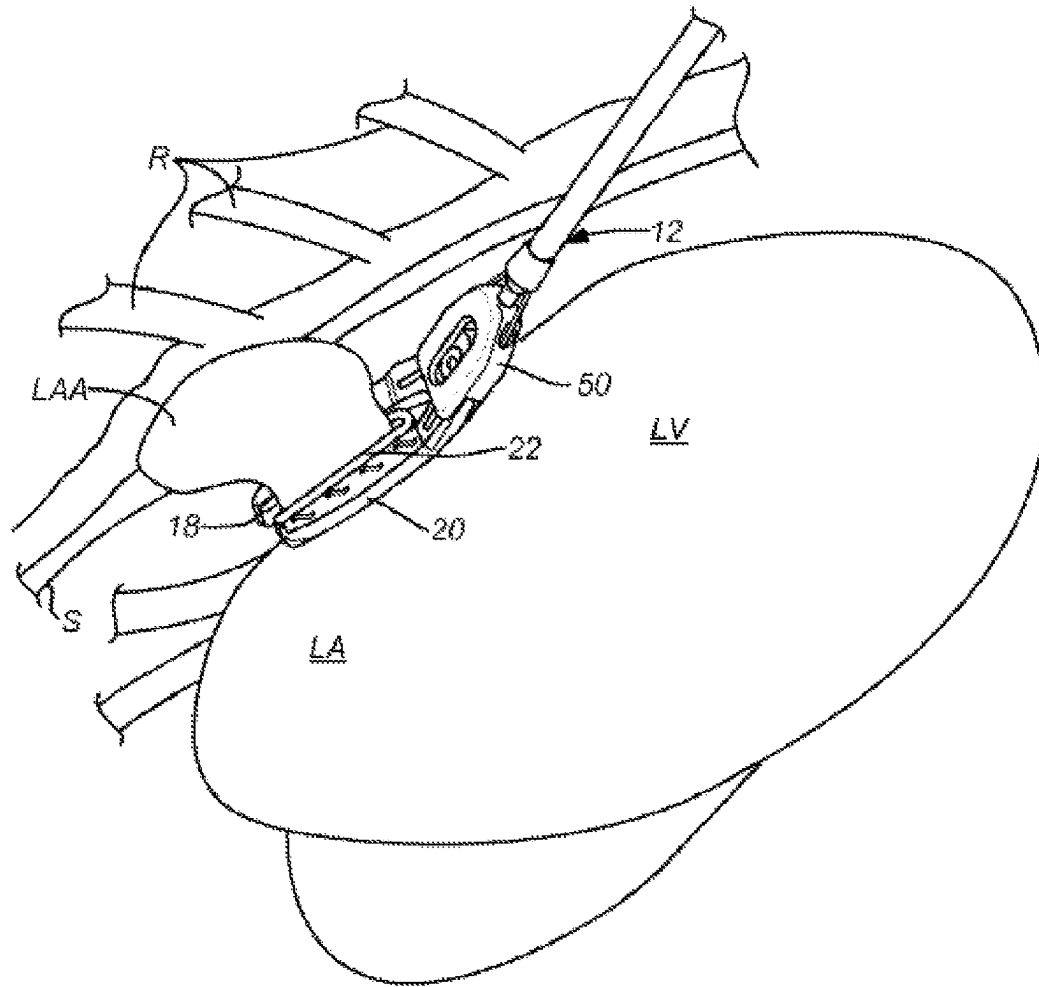


FIG. 11

**FASTENER APPLICATOR WITH
INTERLOCK****RELATED APPLICATIONS**

This application is a continuation of U.S. patent application Ser. No. 16/599,779, filed Oct. 11, 2019, which is a divisional of U.S. patent application Ser. No. 15/037,963, filed May 19, 2016, now U.S. Pat. No. 10,485,545, which is a national phase application of International Patent Application PCT/US2014/066438, filed Nov. 19, 2014, which claims the benefit of priority to provisional U.S. Patent Application Ser. No. 61/906,290 filed Nov. 19, 2013. The disclosures of U.S. patent application Ser. No. 16/599,779, U.S. patent application Ser. No. 15/037,963, International Patent Application PCT/US2014/066438, and provisional U.S. Patent Application Ser. No. 61/906,290 are hereby incorporated by reference in their entireties.

FIELD OF THE DISCLOSURE

The present disclosure pertains broadly to the field of fasteners and/or applicators. More specifically, the disclosure relates to surgical applicators of implants and/or fasteners, including but not limited to sterilized fasteners such as staples.

BACKGROUND

Atrial fibrillation is a relatively common condition characterized by a very rapid heartbeat of the left and right atrium. While atrial fibrillation is not normally fatal itself, it has been associated with an increased risk of stroke. It is believed that the rapid heartbeat causes blood to pool in the left atrial appendage which causes emboli that are released into the left atrium from where they can enter the cerebral vasculature, thus causing a stroke. In addition to stroke, the emboli can enter coronary circulation, potentially causing myocardial infarction, or can enter peripheral circulation, potentially causing peripheral vascular disease.

The risk of stroke in patients suffering from atrial fibrillation can be reduced in a variety of ways. For example, blood thinning drugs can be used to reduce the risk of clot formation. The use of blood thinners, however, is contraindicated in patients at risk of bleeding disorders. More aggressive treatment protocols have been proposed which involve closing the left atrial appendage. Closure and excision may be performed in open surgical procedures, typically requiring the patient to be placed on bypass and the chest to be opened through the sternum. Alternatively, thoroscopic and other less invasive procedures have been proposed. U.S. Pat. No. 5,306,234 teaches the performance of heating heart procedures using otherwise conventional surgical techniques. The use of conventional techniques through small chest penetrations while the heart is beating can be difficult to perform. U.S. Pat. No. 5,865,791 describes an intravascular approach where tools are introduced through the vasculature and passed into the left atrium. The tools are used to ablate or fuse the left atrial appendage from the inside using energy, adhesives, or the like. The '791 patent also describes a thoroscopic procedure where a tether is placed over the neck of the atrial appendage and tied off to achieve isolation. The '791 patent still further suggests other closure elements including sutures, staples, shape-memory wires, biocompatible adhesives, and the like. U.S. Pat. No. 6,488,689 describes a transpericardial procedure where the distal tip of the left atrial appendage is grasped

and pulled backwardly through a capture loop which encircles the base of the left atrial appendage.

A compliant closure structure for the sealing bodily structures such as the left atrial appendage is described in co-pending, commonly owned U.S. Patent Publication 2007/17260278 (application Ser. No. 11/744,135), the full disclosure of which is incorporated herein by reference in its entirety. The compliant structure described in the '278 publication comprises an elastomeric body having a pair of opposed legs which may be arranged in an oval or a U-shaped configuration to define an opening therebetween. By placing the opening between the legs over the left atrial appendage and aligning it with the base of the appendage, the structure may be closed to provide the desired sealing. To hold the structure closed, a number of discrete, axially spaced-apart tissue penetrating fasteners are arranged along the lengths of each of the legs. By compressing the legs together to press-fit the closure devices, the compliant structure may be closed to provide a compliant seal which effectively isolates the left atrial appendage.

The '278 publication describes a particular delivery tool for the compliant closure structure. The delivery tool includes jaws which can be inserted into the legs of the closure structure and actuated to close the jaws in the legs over the left atrial appendage. The jaws further include comb studs which engage and press fit the closure devices in order to hold the compliant structure in its closed, sealing configuration. The studs are intended to be retracted to allow the delivery tool to be removed.

Although functional, the delivery tool of the '278 publication has certain shortcomings. For example, the actuation of the jaws and retraction of the comb studs can be performed out of order, increasing the risk that the delivery of the compliant structure will fail. Moreover, positioning and orientation of the delivery tool can be difficult, particularly when the tool is introduced through an intercostal penetration to access the left atrium. Additionally, the jaws in the device of the '278 publication are attached in the axial plane of the device shall. Such a straight line of attachment can make it more difficult to align the jaws with the base of the appendage and across the os (i.e., ostium) of the atrium leading into the appendage. If the closure device is not aligned across the base to completely close the os, gaps or openings (referred to as "cul-de-sacs") can remain at the site of closure, increasing the risk of thrombus formation in the atrium. The importance of forming a complete seal of the os which is free from such cul-de-sacs is discussed in Sulzberg et al. (2008) Eur. J. Cardiothoracic Surg. 34:766-770.

For these reasons, it would be desirable to provide improved delivery tools for use with the tissue closure devices described in U.S. Patent Publication 2007/0260278. It would be further desirable if the delivery tools and methods of their use were compatible with the delivery of other tissue closure devices and for procedures in addition to closure of the left atrial appendage.

SUMMARY

A surgical applicator for a fastener according to an example embodiment comprises a handle, a first trigger coupled movably with respect to the handle, and a second trigger coupled movably with respect to the first trigger, but operatively arranged to move in tandem with the first trigger during movement of the first trigger. An interlock is coupled to the second trigger and operatively arranged to selectively

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prevent movement of the second trigger relative to the first trigger depending on a position of the first trigger with respect to the handle.

According to an example embodiment, the interlock permits movement of the second trigger relative to the first trigger when the first trigger is moved from a first position to a second position, and restricts movement of the second trigger relative to the first trigger when the first trigger is not in the second position.

According to an example embodiment, a second interlock is coupled to the first trigger and operatively arranged to selectively prevent movement of the first trigger relative to the handle depending on a position of second trigger.

According to an example embodiment, a jaw assembly is selectively opened and closed via movement of the first trigger.

According to an example embodiment, movement of the second trigger selectively retracts one or more fastener supporting structures of the jaw assembly with respect to at least one jaw of the jaw assembly.

According to an example embodiment, the interlock includes a cam follower engaged with a cam channel.

According to an example embodiment, the handle includes the cam channel and the second trigger includes the cam follower.

According to an example embodiment, the cam channel has a first leg that permits tandem movement of the second trigger with the first trigger during closing of the first trigger and a shoulder that prevents movement of the second trigger relative to the first trigger until the first trigger is closed.

According to an example embodiment, the cam channel has a second leg that permits tandem movement of the second trigger with the first trigger during re-opening of the first trigger after the second trigger has been closed relative to the first trigger.

According to an example embodiment, the cam channel includes at least one leg formed concentrically with respect to a first pivot about which the first trigger is rotatably coupled to the handle and a transverse portion funned concentrically with respect to a second pivot about which the second trigger is rotatably coupled to the first trigger.

According to an example embodiment, the at least one leg includes two legs, and the transverse portion is connected between the two legs.

A fastener applicator according to an example embodiment comprises a handle, a first trigger coupled rotatably with respect to the handle, and a second trigger coupled rotatably with respect to the handle. An interlock is disposed with the second trigger and comprises a cam follower engaged in a cam channel that selectively prevents rotation of the second trigger relative to the first trigger depending on a location of the cam follower within the cam channel. The location of the cam follower in the cam channel is set by a position of the first trigger.

According to an example embodiment, the handle includes the cam channel and the second trigger includes the cam follower.

According to an example embodiment, the cam channel has at first leg that permits tandem movement of the second trigger with the first trigger during closing of the first trigger and a shoulder that prevents movement of the second trigger relative to the first trigger until the first trigger is closed.

According to an example embodiment, the cam channel has a second leg that permits tandem movement of the second trigger with the first trigger during re-opening of the first trigger after the so and trigger has been closed relative to the first trigger.

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A fastener applicator according to an example embodiment comprises a handle, a shaft extending distally from the handle and a jaw assembly at a distal end of the shaft having one or more fastener supporting structures. A first trigger is coupled movably with respect to the handle and to the jaw assembly such that movement of the first trigger relative to the handle selectively closes the jaw assembly. A second trigger is coupled movably with respect to the first trigger, the second trigger coupled to the jaw assembly such that movement of the second trigger relative to the first trigger selectively retracts die one or more fastener supporting structures; and an interlock coupled to the second trigger and operatively arranged to selectively prevent movement of the second trigger relative to the first trigger depending on a position of the first trigger.

According to an example embodiment, the interlock permits movement of the second trigger relative to the first trigger when the first trigger is closed and prevents movement of the second trigger relative to the first trigger when the first trigger is open.

According to an example embodiment, a second interlock is coupled to the first trigger and operatively arranged to selectively prevent movement of the first trigger relative to the handle depending on a position of second trigger.

According to an example embodiment, the interlock includes a cam follower engaged with a cam channel.

According to an example embodiment, the handle includes the cam channel and the second trigger includes the cam follower.

According to an example embodiment, the second trigger is movable in tandem with the first trigger during movement of the first trigger.

A fastener applicator according to an example embodiment comprises a handle, a first trigger movable with respect to the handle between a first position and a second position, and a second trigger movable with respect to the first trigger between an initial position and an actuated position. A first interlock is coupled to the second trigger and operatively arranged to selectively prevent movement of the second trigger relative to the first trigger until the first trigger is moved into the second position. A second interlock is coupled to the first trigger and operatively unarmed to selectively prevent movement of the first trigger relative to the handle until the second trigger is moved into the actuated position.

According to an example embodiment, a pair of jaws is included, with the first trigger coupled to the pair of jaws for setting the pair of jaws in an open configuration when the first trigger is in the first position and in a closed configuration when the first trigger is in the second position.

According to an example embodiment, one or more fastener supporting structures are disposed with the pair of jaws. The second trigger is coupled to the one or more fastener supporting structures for setting the one or more fastener supporting structures in a deployed configuration when the second trigger is in the initial position and in a retracted configuration when the second trigger is in the actuated position.

A method of operating a fastener applicator according to an example embodiment comprises: (i) repositioning a first trigger of the fastener applicator with respect to a handle from a first position to a second position, (ii) moving a second trigger in tandem with the first trigger, (iii) preventing repositioning of a second trigger between an initial position and an actuated position relative to the first trigger with an interlock coupled to the second trigger until the first trigger is moved to the second position, and (iv) reposition-

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ing the second trigger to the actuated position relative to the first trigger after the first trigger is moved to the second position.

A method of operating a fastener applicator according to an example embodiment comprises: (i) moving a first trigger of the fastener applicator with respect to a handle from a first position to a second position, (ii) preventing movement of a second trigger between an initial position and an actuated position relative to the first trigger with an interlock coupled to the second trigger until the first trigger is moved to the second position, (iii) rearranging a cam follower and a cam channel of the interlock with respect to each other due to the moving of the first trigger to the second position, and (iv) moving the second trigger to the actuated position relative to the first trigger after the first trigger is moved to the second position.

A method of operating a fastener applicator according to an example embodiment comprises: (i) moving a first trigger of the fastener applicator with respect to a handle from a first position to a second position, (ii) preventing movement of a second trigger between an initial position and an actuated position relative to the first trigger with a first interlock coupled to the second trigger until the first trigger is moved to the second position, (iii) moving the second trigger to the actuated position relative to the first trigger after the first trigger is moved to the second position, (iv) preventing movement of the first trigger relative to the handle back to the first position with a second interlock coupled to the first trigger until the second trigger is moved to the actuated position, and (v) moving the first trigger back to the first position relative to the handle after the second trigger is moved to the actuated position.

A method of operating a fastener applicator according to an example embodiment comprises: (i) repositioning a first trigger of the fastener applicator with respect to a handle from a first position to a second position, (ii) closing a jaw assembly coupled to the handle of the fastener applicator due to movement of the first trigger to the second position, (iii) preventing repositioning of a second trigger between an initial position and an actuated position relative to the first trigger with an interlock coupled to the second trigger until the first trigger is moved to the second position, (iv) repositioning the second trigger to the actuated position relative to the first trigger after the first trigger is moved to the second position, and (v) retracting one or more fastener supporting structures comprising at least one protrusion movably extendable and retractable with respect to at least one jaw of the jaw assembly of the fastener applicator due to repositioning of the second trigger to the actuated position.

According to an example embodiment, the fastener applicator comprises a second interlock coupled to the first trigger and the method further comprises preventing movement of the first trigger from the second position to the first position with the second interlock until the second trigger is moved to the actuated position relative to the first trigger, and repositioning the first trigger from the second position back to the first position after the second trigger is moved to the actuated position relative to the first trigger.

According to an example embodiment, the interlock includes a cam follower engaged with a cam channel.

According to an example embodiment, rotating the first trigger from the first position to the second position includes traversing the cam follower along a first leg of the channel.

According to an example embodiment, the first leg is formed concentrically with respect to a pivot about which the first trigger rotates.

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According to an example embodiment, rotating the second trigger to the actuated position relative to the first trigger includes traversing the cam follower along a portion of the channel transverse to the first leg.

According to an example embodiment, the handle includes the cam channel and the second trigger includes the cam follower.

According to an example embodiment, rotating the first trigger from the first position to the second position including moving the second trigger in tandem with the first trigger.

BRIEF DESCRIPTION OF THE DRAWINGS

The following descriptions should not be considered limiting in any way. With reference to the accompanying drawings, like elements are numbered alike:

FIG. 1 is a perspective view of a closure device applicator constructed in accordance with to an example embodiment of the present disclosure;

FIG. 2 is a partial cross-sectional elevation view of the closure device applicator of FIG. 1, shown with the jaws open and rotated 90° relative to the position shown in FIG. 1;

FIG. 3 is a partial, cross-sectional elevation view of the device applicator similar to FIG. 2, shown with the first trigger closed relative to the handle in order to close the jaws, and the second trigger open (or spaced apart) with respect to the first trigger;

FIG. 4 is a partial, cross-sectional elevation view of the closure device applicator similar to that shown in FIGS. 2 and 3 with the second trigger closed relative to the first trigger demonstrating a retraction of the closure device-engaging studs;

FIG. 5 is a partial, cross-sectional elevation view of the closure device applicator of FIGS. 2-4, shown with the first trigger re-opened with respect to the handle in order to open the jaws while the closure device-engaging studs remain retracted;

FIG. 6A-6D illustrate the movement of the jaw and stud engagement with the closure devices of a tissue closure device according to an example embodiment of the present disclosure;

FIG. 7 illustrates an exemplary mechanism by which the closure device-engaging studs are retracted within the jaws according to an example embodiment of the present disclosure;

FIG. 8 illustrates an interlock formed between a cam channel, slot, or groove in a handle of the applicator of FIG. 1, shown in cross-section, and a cam follower projecting from a second trigger of the applicator of FIG. 1, shown in a perspective and exploded view orientation according to an embodiment of the present disclosure;

FIGS. 9A-9E illustrate various positions of the cam follower of FIG. 4 along the cam channel of the interlock as first and second triggers of the applicator are moved through the configurations depicted in FIGS. 2-5;

FIG. 10 illustrates operation of the interlock of FIG. 8 with respect to the first and second triggers of the applicator of FIG. 1; and

FIG. 11 illustrates use of the applicator according to an example embodiment of the present disclosure, wherein the applicator is used for closing a closure device over a left atrial appendage.

DETAILED DESCRIPTION

A detailed description of one or more embodiments of the disclosed apparatus and method are presented herein by way of exemplification and not limitation with reference to the Figures.

Embodiments of the present disclosure provide alternative and improved apparatuses, systems, and methods for deploying one or more closure devices or fasteners to tissue. An exemplary tissue structure of a patient at risk of stroke or other adverse events resulting from emboli released into circulation from the left atrial appendage, may include the left atrial appendage. Patients benefiting from the exemplary procedures disclosed herein will at least partially include those suffering from atrial fibrillation which can cause clot and thrombus formation in the left atrial appendage, thus increasing the chance of emboli release.

Exemplary embodiments of the present disclosure provide a mechanical closure device applicator for introducing the fastener over the tissue structure, which fastener is left in place in order to close and/or seal the tissue structure. A portion of the tissue structure extending beyond the fastener may then be cut, excised, or otherwise removed, although this may be left to the physician's preference. In one embodiment, the tissue fastener comprises a compression body having at least two opposed, compliant tissue-engaging surfaces which are placed over opposite sides of the tissue structure. In this embodiment, the tissue-engaging surfaces are held together by a plurality of axially spaced-apart tissue-penetrating fasteners, which extend from one of the surfaces, through the intermediate tissue, and into the other surface to both hold the compression body in place and to apply a desired level of compression force, which is determined by both the softness of the compression body and the distance between the surfaces when they are fully attached. A well may be provided in the compression body around the tissue-penetrating barb of the fastener such that a gasket seal is formed by the compression body around the puncture site in the tissue. A stabilizing lip may be provided in one leg of the compression body to prevent a rolling motion of one leg with respect to another leg of the compression body in order to keep the two opposing soft members linearly aligned. More detailed descriptions of fasteners suitable for use with embodiments of the present disclosure are found in patent publication U.S. 2007/0260278, the full disclosure of which has been previously incorporated herein by reference.

Referring now to FIG. 1, a fastener applicator 10 constructed in accordance with various embodiments of the present disclosure comprises a shaft 12 having a handle assembly 14 at its proximal end, and a jaw assembly 16 at a distal end, when individual jaws 18 and 20 are adapted to carry a fastener 22, as best illustrated in FIGS. 6A-6D. The fastener 22 is shown in an example embodiment to have a U-shaped configuration which defines a V-shaped region 24 for receiving the left atrial appendage or other tissue structure when the jaws are open as shown in FIGS. 1, 2, and 6A. A plurality of studs 26 are formed along the inner surfaces of each jaw 20 and 18 to engage tissue-penetrating fasteners which comprise penetrating components 28 and receptacle components 30, as best shown in FIG. 6A. The handle assembly 14 will include a handle 32, a first trigger 34, and a second trigger 36.

Referring now to FIGS. 2 and 6A, the fastener applicator 10 is shown in its shelf or delivery configuration with the fastener 22 received over the jaw assembly 16 and the V-shaped opening 24 in the device ready to be placed over

a tissue structure such as the left atrial appendage (see FIG. 10). The triggers 34 and 36 are each respectively movable between an initial or first position and an actuated or second position. As will be discussed in more detail below, the first trigger 34 is movable between its respective positions relative to the handle 32, while the second trigger 36 is movable between its respective positions relative to the first trigger 34. In one embodiment, in their respective first positions, the first trigger 34 and the second trigger 36 are initially in a fully open position, that is, pivoted fully away from the handle 32. For this reason, it is to be understood that any mention of "opening" the first trigger 34 refers more generally to moving the first trigger 34 toward its first or initial position, while any mention of "closing" the first trigger 34 or the second trigger 36 refers more generally to moving the first trigger 34 or the second trigger 36 towards its respective second or actuated position.

After the fastener 22 is advanced over the left atrial appendage or other target tissue structure, the jaws 18 and 20 are closed by manually pulling the first trigger 34 toward the handle 32, as shown in FIGS. 3 and 6B. The first trigger 34 is mounted on a pin or pivot 40, which is shown to be fixedly secured to the handle assembly 14, thereby rotatably coupling the first trigger 34 to the handle 32. Closing the first trigger 34 with respect to the handle 32, that is, pivoting about the pivot 40 from the first position of the first trigger 34 to the second position, proximally retracts the rod 42 which is linked to the first trigger 34 by a pin or pivot 41 and a lever 43. The rod 42 is attached to a pin 44 (FIG. 6B) which is pivotally attached to the proximal ends of the jaws 18 and 20 and proximally retracts the jaws so that they are closed by the movement of slots 46 over pins 48 in an end frame 50 attached to a stationary sleeve 52. When the jaws 18 and 20 are closed, the studs 26 engage the tissue penetrating components 28 and the tissue penetrating receptacles 30 so that they engage and lock with each other, thus closing the two legs of the fastener 22, as shown in FIG. 6B.

The second trigger 36 may be mounted on the pin 41 so that the second trigger 36 moves, actuates, or closes with respect to the handle 32 in tandem with the first trigger 34. In this way, the second trigger 36 is rotatably coupled to the handle 32, although indirectly via the pin 41 and the first trigger 34. By "in tandem" it is meant that movement of the first trigger 34 translates at least partially into corresponding movement of all or portions of the second trigger 36. In the illustrated embodiment, movement of the first trigger 34 causes a substantially equivalent amount of rotational movement of the second trigger 36, although there could be gears, linkages, springs, or the like to cause a lesser or greater degree of movement of the second trigger 36 to result from corresponding movement of the first trigger 34.

The applicator 10 may optionally include an interlock 60 in some embodiments. According to the illustrated embodiment, as the first trigger 34 and the second trigger 36 are moved in tandem, a ratcheting tooth surface 62 of the interlock 60 is closed against pins 64, as shown in FIG. 3, so that the first trigger 34 cannot be re-opened once the jaw assembly 16 has been closed, either completely or to a certain degree of closure or spacing between the jaws. This is advantageous as described above, since the jaws should not be completely opened prior to retracting the studs 26 by closing the second trigger 36, as will be described below. The ratcheting tooth surface 62 can include any number of ratchets or notches such that the first trigger 34 can be progressively closed (i.e., brought into approximation of the handle 32) by more fully driving the ratcheting tooth surface

62 along the pins 64 in the direction of closure for the first trigger 34 (e.g., from the configuration of FIG. 2 to the configuration of FIG. 3).

Additionally, as the first trigger 34 is moved to its actuated, closed, or second position (i.e., moved or actuated to the configuration of FIG. 3) in order to close the jaws 18 and 20 of the jaw assembly 16, a corner 66 (see FIG. 2) of the interlock 60 will engage an inclined surface 68 on a clicker pin 70 to disengage the clicker pin 70 from a holding pin 72, thus allowing a spring 74 to push the pin 70 downward so that a bottom surface 76 thereof strikes the bottom of the handle 32, thus causing a loud click to audibly alert the physician that the jaws have been closed and the interlock 60 has been engaged. This audible confirmation indicates to the physician or other medical personnel that the fastener 22 has been closed and that the studs 26 can be retracted.

The studs 26 are initially in an extended configuration, as shown in FIGS. 2-3 and 6A-6B, and then retracted by closing second trigger 36 against the first trigger 34, as shown in FIGS. 4 and 6C. Closure of the second trigger 36 relative to the first trigger 34 draws the outer sleeve 54 proximally over the stationary sleeve 52 which draws pull wires 80 proximally to full stud combs 84 or other fastener supporting structures, proximally, as shown in FIG. 7. The stud combs 84 are mounted on pins 86 which travel in slots 88 formed in each of the jaws 18 and 20. The second trigger 36 is pivotally mounted on, or rotatably coupled to, the first trigger 34 by the pin 41 and is coupled to the stationary sleeve by pins 90 and 91 and a lever 92. It is to be appreciated that the studs 26 can take other shapes or arrangements and/or that other fastener supporting structure (i.e., structures capable of supporting the fasteners 22 during closing and engaging thereof) can be included in lieu of the studs 26 that are either disengaged from or engaged to the fastener 22 upon intentional closure or movement of the second trigger 36. For example, in lieu of a plurality of individual studs 26, a supporting structure in the form of a single unitary rail supporting all of the components 28 can be included in one embodiment.

Movement of the second trigger 36 relative to the first trigger 34, e.g., closure of the second trigger 36, disengages the interlock 60 by engaging a ramped surface 94 attached to the interlock 60 against a fixed pin 96 in the second trigger 36. In this way, the interlock 60 selectively permits or prevents movement, e.g., opening, of the first trigger 34 depending on a position of the second trigger 36. In other words, movement of the second trigger 36 to its corresponding second position, relative to the first trigger 34, releases the interlock 60. The pin 96 follows the path shown in broken line in FIG. 3 to engage the surface 94. The ramped surface 94 is caused to move downwardly, pulling the ratchet teeth 62 away from the fixed pins 64, as shown in FIGS. 4 and 5. The triggers 34 and 36 are now ready to be opened in order to open the jaws, as will now be described.

Referring now to FIGS. 5 and 7C, the jaws 18 and 20 may be opened by releasing manual compression on the first and second triggers 34 and 36, allowing spring 98 which was compressed during closure of the first trigger 34 to push rod 42 (attached to pivot 44) distally forward, opening the jaws and leaving the fastener 22 in place.

FIG. 8 shows a cross-section of the handle 32 as well as the second trigger 36 exploded therefrom. An interlock 100 is included between the handle 32 and the second trigger 36 to selectively permit and prevent rotational movement of the second trigger 36 with respect to the first trigger 34. The first trigger 34 does not form a part of the interlock 100 in the

illustrated embodiment, and therefore is not illustrated in FIG. 8. However, it is to be understood that the first trigger 34 in the embodiment of FIG. 8 would be nevertheless be secured to the second trigger 36 at the pivot 41 as described herein with respect to the other Figures. Additionally, the shaft 12 and the jaw assembly 16 are not illustrated in some Figures, including FIG. 8, but can be included as described herein with respect to the other Figures.

More specifically, the interlock 100 includes a cam follower 102 and a channel 104. In the illustrated embodiment, the cam follower 102 is formed as a protrusion or projection extending from or otherwise coupled to the second trigger 36, and is preferably round in cross section but alternatively may be of other shapes. The cam follower 102 can be integrally formed with the second trigger 36 or affixed thereto as a separate component. The channel 104 may be formed as a groove, depression, notch, slot, or the like in the handle 32. The follower 102 is configured to engage with and/or travel along the channel 104 (and thus will be restricted to movement within the channel 104) when the second trigger 36 is assembled with the handle 32. It is noted that the applicator 10 can include other exemplary embodiments of a single follower 102 and a single channel 104 if desired, but for symmetry, balance, etc., the second trigger 36 may include two cam followers 102 (e.g., another follower 102 hidden from view in FIG. 8 on the opposite lateral side of the second trigger 36 and mirroring the illustrated one of the followers 102 about a plane bisecting the second trigger 36), and the handle 32 may correspondingly include two cam grooves 104 (e.g., another channel 104 located in the half of the handle 32 not shown in FIG. 8, which mirrors the illustrated channel 104 about the plane used to form the cross-sectional view of the handle 32 in FIG. 8).

The mechanical coupling of the follower 102 to the second trigger 36 prevents rotation of the second trigger 36 relative to the first trigger 34 when the follower 102 abuts against the walls or shoulders defining the channel 104. For example, FIGS. 9A-9E show various positions of the follower 102 (without the second trigger 36 for clarity) corresponding to the various combinations of positions of the first and second triggers 34 and 36 of the applicator 10. That is, for example, the position of the follower 102 in FIG. 9A corresponds to an initial, shelf, or delivery configuration of the applicator 10 shown in FIG. 2, i.e., before the triggers 34 or 36 have been actuated.

In the position of FIG. 9A, rotation of the second trigger 36 is prevented by interference of the follower 102 with a shoulder 106 of the channel 104. That is, movement of the second trigger 36 relative to the first trigger 34 occurs about the pin 41, which rotationally couples the second trigger 36 to the first trigger 34. Thus, rotation of the second trigger 36 about the pin 41 (if unblocked) would result in the cam follower 102 following an arc defined by a circular path concentric with the pin 41, e.g., illustrated as a circle 107 in FIG. 10. Front FIG. 10, it is clear that rotational movement of the cam follower 102 along an arc defined by the circle 107 is not possible due to interference between the cam follower 102 and the shoulder 106. Accordingly, rotation of the second trigger 36 about the pin 41 relative to the first trigger 34 is prevented by the interference between the cam follower 102 and the shoulder 106.

The channel 104 is shaped with a leg 108, which permits the second trigger 36 to travel in tandem with the first trigger 34 as the first trigger 34 is moved toward its second position, i.e., rotated about the pivot 40 toward the configuration of the applicator 10 shown in FIG. 3. That is, again referring to FIG. 10, tandem movement of second trigger 36 due to

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movement of the first trigger **34** would result in the cam follower **102** traveling along an arc defined by a circular path concentric with the pivot **40**, illustrated in FIG. **10** by a circle **109**. The first leg **108** is arranged to lie substantially along such an arc formed by the circle **109**, thereby enabling the cam follower **102** to travel along the leg **108** during rotation of the first anti second triggers **34** and **36** about the pin **40**.

FIG. **4B** illustrates the position of the follower **102** in the channel **104** when the first trigger **34** is rotated partially toward the configuration of FIG. **3**, e.g., with ratcheting tooth surface **62** of the interlock **60** at most only partially driven along and engaged with the pins **64**. At the position of FIG. **9B**, the shoulder **106** continues to prevent rotation of the second trigger **36** by blocking movement of the follower **104** in the direction necessary to rotate the second trigger **36**.

FIG. **9C** illustrates the position of the follower **102** after the trigger first **34** has been moved to the configuration of FIG. **3** and the ratcheting tooth surface **62** of the interlock **60** is fully driven along and engaged with the pins **64**. It can be appreciated in view of FIGS. **3** and **9C** that rotation of the second trigger **36** becomes possible at this position because the shoulder **106** no longer abuts the follower **102**. Alternatively stated, it is to be appreciated that tandem movement of the second trigger **36** with the first trigger **34** has shifted the position of the cam follower **102** and the pin **41** relative to the channel **104** (since the pin **41** and the cam follower **102** are directly coupled to the second trigger **36** in the illustrated embodiment). It is to be further understood than this shifting of the position of the pin **41** and the cam follower **102** has also moved the location of the circle **107** to align with a portion **110** of the channel **104** that is transverse to the leg **108** and arranged along an arc that is concentric with respect to the pin **41**.

In view of the foregoing it can be seen that the position of the first trigger **34** selectively prevents or permits rotation of the second trigger **36**. Again, movement of the first trigger **34** causes tandem movement of the second trigger **36**, which results in the follower **102** first traversing along the leg **108** of the channel **104**. However, movement of the second trigger **36** relative to the first trigger **34** is prevented due to interference between the cam follower **102** and the shoulder **106**. Relative movement of the second trigger **36** with respect to the first trigger **34** is thus prevented until the cam follower **102** is moved to a position at which the cam follower **102** no longer abuts the shoulder **106**. FIG. **9D** shows the position of the follower **102** after movement of the second trigger **36** relative to the first trigger **34** is permitted and the second trigger **36** has also been actuated to its second position, thereby corresponding to the configuration of the applicator **10** shown in FIG. **4**.

The channel **104** is shaped with another leg **112** that enables the follower **102** to continue to traverse the channel **104** such that the second trigger **36** can again move in tandem with the first trigger **34** as the first trigger **34** is moved to reopen the jaw assembly **16**, i.e., as the first trigger **34** is moved back toward its first position. It is to be appreciated that the leg **112**, similar to the leg **108**, is formed along an arc that is concentric to the pivot **40**. This accordingly enables the cam follower **102** to travel along the leg **112** as the second trigger **36** moved in tandem with the first trigger **34** as the first trigger **34** is rotated about the pivot **40**. FIG. **9E** accordingly shows the corresponding location of the follower **102** in the channel **104** when the first trigger **34** is returned to its first or initial position, which may correspond to the applicator **10** assuming the configuration of FIG. **5** in which the jaw assembly **16** is again opened.

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It is to be appreciated that the channel **104** can take other shapes in other embodiments depending on the desired relative movement of the first and second triggers **34** and **36** relative to each other and to the handle **32**. More specifically, the channel **104** will include ii leg or portion for each optionally desired stage of movement of the first trigger **34** and of the second trigger **36**, with the legs and/or portions of the channel **104** being concentric with respect to the pivot about which rotation is desired for each stage of movement. Thus, in order to sequentially permit (i) tandem movement of both of the first and second triggers **34** and **36** about a first pivot (e.g., the pivot **40**), then (ii) movement of the second trigger **36** relative to the first trigger **34** about a second pivot (e.g., the pivot **41**), and then (iii) tandem movement of the first and second triggers **34** and **36** again with respect to the first pivot, the channel **104** will include two legs (e.g., the legs **108** and **112**) that are spaced apart but both concentric with respect to the first pivot (e.g., the pivot **40**), which enables the movements of stages (i) and (iii), and a transverse portion (e.g., the portion **110**) connecting between the two legs (e.g., the legs **108** and **112**) that is formed concentrically with respect to the second pivot (e.g., the pivot **41**) to enable the movement of stage (ii).

From the above description and the FIGS. **9A-9E**, it is clear that the interlock **100** accordingly prevents rotation of the second trigger **36** until after the trigger first **34** is first rotated, preferably fully rotated (i.e., from a first position to a second position). Since full rotation of the first trigger **34** (e.g., as shown in FIG. **3**) corresponds to the closing of the jaw assembly **16**, the interlock **100** thus advantageously selectively prevents retraction of the studs **26** (due to rotation of the second trigger **36**) until after the jaw assembly **16** is closed. Referring again to the above description, it is again noted that the interlock **60** is released depending on the position of the second trigger **36** relative to the first trigger **34**. Thus, the location of the first trigger **34** is first used to release the interlock **100**, selectively permitting movement of the second trigger **36** relative to the first trigger **34**, and thereafter the location of the second trigger **36** relative to the first trigger **34** is used to release the interlock **60**, which selectively permits movement of the first trigger **34** relative to the handle **32** back toward the initial position for the first trigger. In this way, the aforementioned order of the steps of (1) closing the jaws, (2) retracting the studs, and (3) opening the jaws is maintained and ensured by use of either or both of the interlocks **60** and **100**.

It is to be appreciated that the interlock **100** can take other forms in non-illustrated embodiments. For example, in one embodiment the channel **104** is alternatively formed with the second trigger **36** and the follower **102** with the handle **32**. In another embodiment, part of the channel **104** is formed in one of the handle halves while the other is formed in the other of the handle halves.

Referring now to FIG. **11**, one specific example of use is illustrated. In this example, the fastener applicator **10** is used to deliver one of the fasteners **22** over the base of a left atrial appendage LAA in an open chest, beating heart procedure. The sternum **S** is opened, spreading the ribs **R** to provide a working space over the heart. After opening the pericardial sack, the heart may be lifted and turned, for example using a sheet of gauze or other material which is placed behind the heart, to expose the left atrial appendage within the opening as shown in FIG. **10**. After the left atrial appendage LAA is exposed, the jaws **18** and **20** of the fastener applicator are placed around the base of the appendage by manipulating the shaft **12**, as shown in FIG. **10**. The angle of the jaws **18** and **20** relative to the shaft, typically about 15°, is highly

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advantageous as it allows the jaws to engage the base of the left atrial appendage so that they are generally parallel with the os between the appendage and the left atrium LA. If the plane of the jaws was aliened with the shaft, it would be difficult to achieve this orientation and the risk of closing the appendage and leaving a cul-de-sac (an open space beyond the us and into interior of the left atrial appendage) is greatly increased. Such cul-de-sac is problematic as it can be a source of clotting and, if created, must be closed in the same or later procedure. Once the jaws **18** and **20** properly position the fastener **22** about the base of the left atrial appendage, the jaws are actuated and the fastener deployed as described previously.

While the invention has been described with reference to an exemplary embodiment or embodiments, it will be understood by those skilled in the art that various changes may be made and equivalents may be substituted for elements thereof without departing from the scope of the invention. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the invention without departing from the essential scope thereof. Therefore, it is intended that the invention not be limited to the particular embodiment disclosed as the best mode contemplated for carrying out this invention, but that the invention will include all embodiments falling within the scope of the claims. Also, in the drawings and the description, there have been disclosed exemplary embodiments of the invention and, although specific terms may have been employed, they are unless otherwise stated used in a generic and descriptive sense only and not for purposes of limitation, the scope of the invention therefore not being so limited. Moreover, the use of the terms first, second, etc. do not denote any order or importance, but rather the terms first, second, etc. are used to distinguish one element from another. Furthermore, the use of the terms a, an, etc. do not denote a limitation of quantity, but rather denote the presence of at least one of the referenced item.

What is claimed is:

1. A method of operating a fastener applicator, the method comprising:
repositioning a first trigger of the fastener applicator with respect to a handle from a first position to a second position;

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closing a jaw assembly coupled to the handle of the fastener applicator due to movement of the first trigger to the second position;

preventing repositioning of a second trigger between an initial position and an actuated position relative to the first trigger by an interlock coupled to the second trigger until the first trigger is moved to the second position, wherein the interlock includes a cam follower engaged with a cam channel, and wherein the handle includes the cam channel and the second trigger includes the cam follower;

repositioning the second trigger to the actuated position relative to the first trigger after the first trigger is moved to the second position; and

retracting one or more fastener supporting structures comprising at least one protrusion movably extendable and retractable with respect to at least one jaw of the jaw assembly of the fastener applicator due to repositioning of the second trigger to the actuated position.

2. The method of claim **1** wherein the fastener applicator comprises a second interlock coupled to the first trigger, the method further comprising preventing movement of the first trigger from the second position to the first position with the second interlock until the second trigger is moved to the actuated position relative to the first trigger, and repositioning the first trigger from the second position back to the first position after the second trigger is moved to the actuated position relative to the first trigger.

3. The method of claim **1**, wherein rotating the first trigger from the first position to the second position includes traversing the cam follower along a first leg of the channel.

4. The method of claim **3**, wherein the first leg is formed concentrically with respect to a pivot about which the first trigger rotates.

5. The method of claim **1**, wherein rotating the second trigger to the actuated position relative to the first trigger includes traversing the cam follower along a portion of the channel transverse to the first leg.

6. The method of claim **1**, wherein rotating the first trigger from the first position to the second position includes moving the second trigger in tandem with the first trigger.

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